

- Se aplica cuando se tiene una regla de asignación

- Ventaja: Validez Interna (Exp. Controlado)

- Desventaja: Validez Externa (LATE)

$$D_i = \begin{cases} 0 \\ 1 \end{cases}$$

$z_i \rightarrow$ Variable de asignación (Var. cont.)

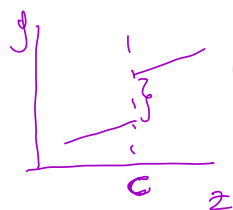
$$\therefore D_i = 1 [\underline{z_i \geq c}] \quad D_i = \begin{cases} 0 & \text{si } z_i < c \\ 1 & \text{si } z_i \geq c \end{cases}$$

$$\therefore D_i = 1 [z_i \leq c] \quad D_i = \begin{cases} 0 & \text{si } z_i > c \\ 1 & \text{si } z_i \leq c \end{cases}$$

Supuesto de identificación.

- Continuidad local.

$$\leadsto y(D_i=1), y(D_i=0) \perp D_i | z_i$$

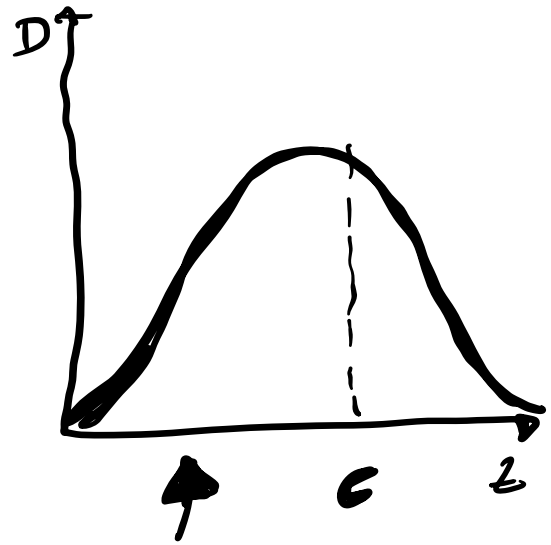
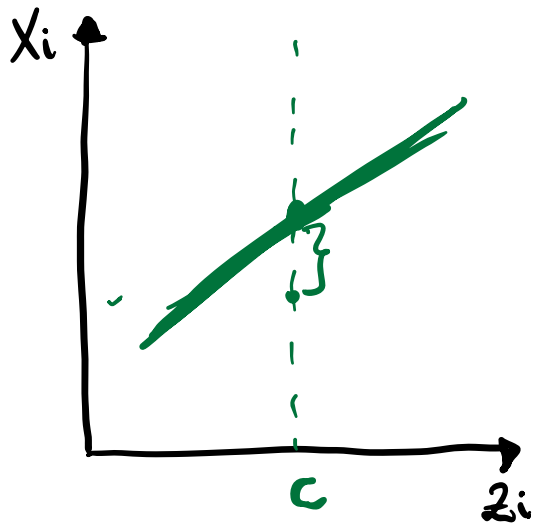
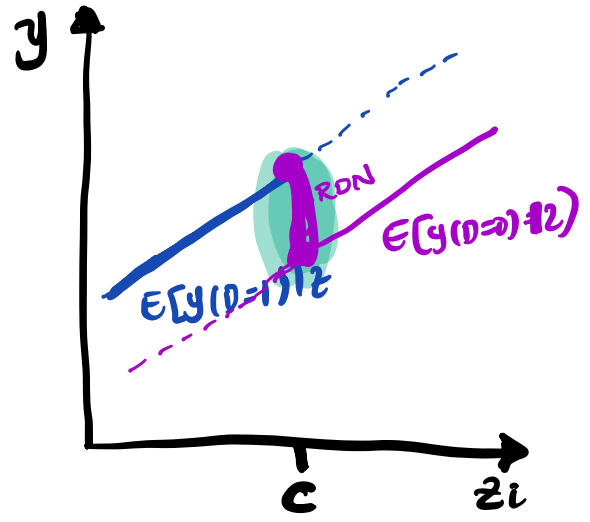
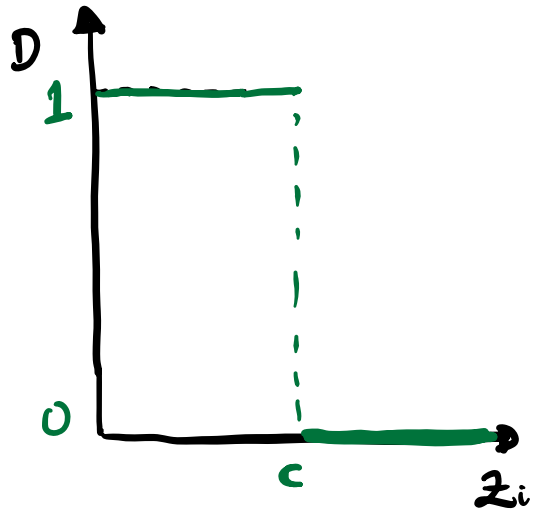


$E[y(D_i=d) | z_i=z]$ es continuo en z cerca de $z_i=c$ para $d=0,1$

$$\tau_{RD} = E[y(D_i=1) - y(D_i=0) | z_i=c]$$

$$= \lim_{z \downarrow c} E[y(D_i=1) | z_i=c] - \lim_{z \uparrow c} E[y(D_i=1) | z_i=c]$$

①



★ Estimación: (fórmula/Receta)



- 1 Recorte la muestra a una ventana razonable alrededor del umbral $z - h \leq z_i < z + h$ donde $h > 0$ $h = \text{ancho de Banda}$.

- 2 Recodifique la variable de asignación en desviaciones del umbral

$$\tilde{z}_i = z_i - c$$

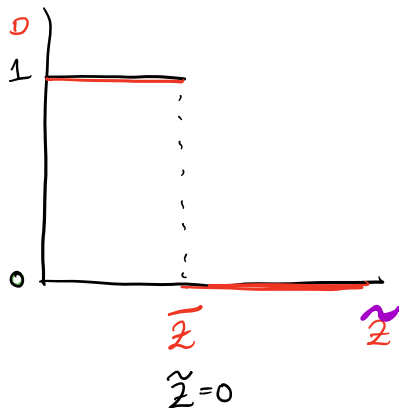
$\tilde{z} = 0$ si está en el umbral

$\tilde{z} > 0$ si $D = 1$ (o $\tilde{z} > 0$ si $D = 0$)

$\tilde{z} < 0$ si $D = 0$ (o $\tilde{z} < 0$ si $D = 1$)

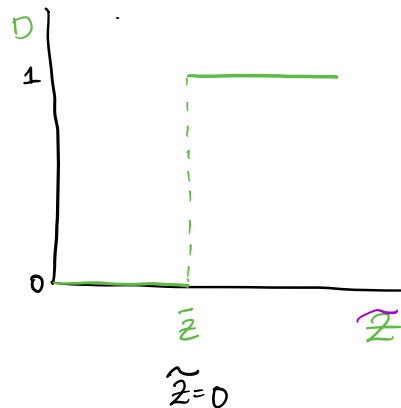
- 3 Decida un modelo para $E[y|x]$.

Lineal / Misma Pendiente para $E[y(D=0)|z]$ y $E[y(D=1)|z]$



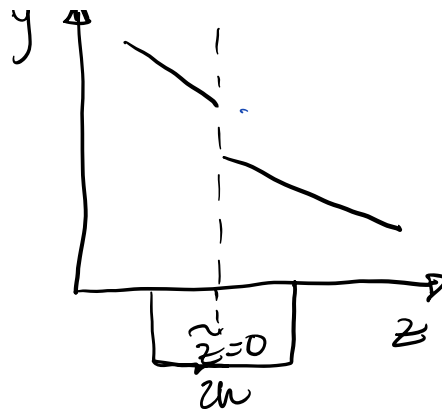
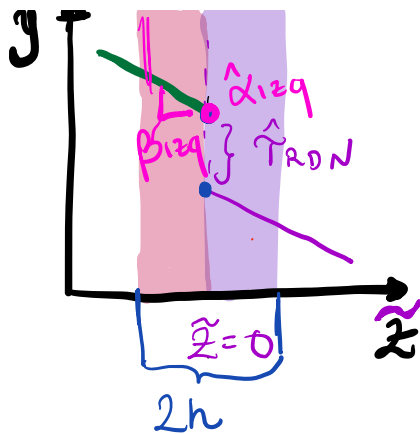
$D = 1$ si $\tilde{z} < 0$

$D = 0$ si $\tilde{z} > 0$



$D = 1$ si $\tilde{z} > 0$

$D = 0$ si $\tilde{z} < 0$



$$\hat{\tau}_{RDN} = \bar{Y}_{12q} - \bar{Y}_{0er}$$

$$\hat{\tau}_{RDN} = \bar{Y}_{0er} - \bar{Y}_{12q}$$

$$\bar{Y}_{12q}$$

$$Y = \alpha_{12q} + \beta_{12q} \tilde{Z} + \epsilon$$

$$E[Y | \tilde{Z} = 0] = \alpha_{12q}$$

$$\bar{Y}_{0er}$$

$$Y = \alpha_{0er} + \beta_{0er} \tilde{Z} + \epsilon$$

$$E[Y | \tilde{Z} = 0] = \alpha_{0er} = \bar{Y}_{0er}$$

$$\hat{\tau}_{RD} = \bar{Y}_{12q} - \bar{Y}_{0er}$$

En regresión lineal

① $E[Y(D=0) | Z]$ es lineal y el efecto del programa, τ , no depende de z

$$E[Y(D=0) | Z] = \delta + \beta Z$$

$$E[Y(D=1) - Y(D=0) | Z] = \tau$$

$$E[Y(D=1) | Z] = \delta + \beta Z + \tau$$

$$\begin{aligned} \rightarrow E[Y(D=1) | Z] &= \tau + \underbrace{E[Y(D=0) | Z]}_{\delta + \beta Z} \\ &= \tau + \delta + \beta Z \quad \checkmark \end{aligned}$$

$$\rightarrow Y = D \cdot Y(D=1) + (1-D) \cdot Y(D=0)$$

$$\rightarrow E[Y | Z, D] = E[D \cdot Y(D=1) + (1-D) Y(D=0) | Z, D]$$

$$= D \cdot E[Y(D=1) | Z] + (1-D) E[Y(D=0) | Z]$$

$$= D \cdot E[Y(D=1) | Z] + E[Y(D=0) | Z] - D E[Y(D=0) | Z]$$

$$= \underbrace{E[Y(D=0) | Z]}_{\delta + \beta Z} + D \cdot \underbrace{E[Y(D=1) - Y(D=0) | Z]}_{D = \tau}$$

$$E[Y | D, Z] = \delta + \beta Z + D \cdot \tau + \beta C - \beta C$$

$$= (\delta + \beta C) + \beta(Z - C) + D \tau$$

$$= \alpha + \beta \tilde{z} + D\tau$$

$$y = E[y|D, z] + \epsilon$$

$$= \alpha + \beta \tilde{z} + D\tau + \epsilon$$

$$E[y|D=1, \tilde{z}=0] = \alpha + \tau \quad E[y|D=0, \tilde{z}=0] = \alpha$$

$$(2) \quad E[y(D=0) | z] \quad y \quad E[y(D=1) | z]$$

son lineales pero dependen de z

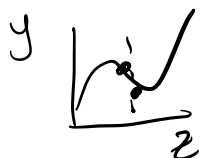
$$E[y(D=0) | z] = \delta_0 + \beta_0 z$$

$$E[y(D=1) | z] = \delta_1 + \beta_1 z$$

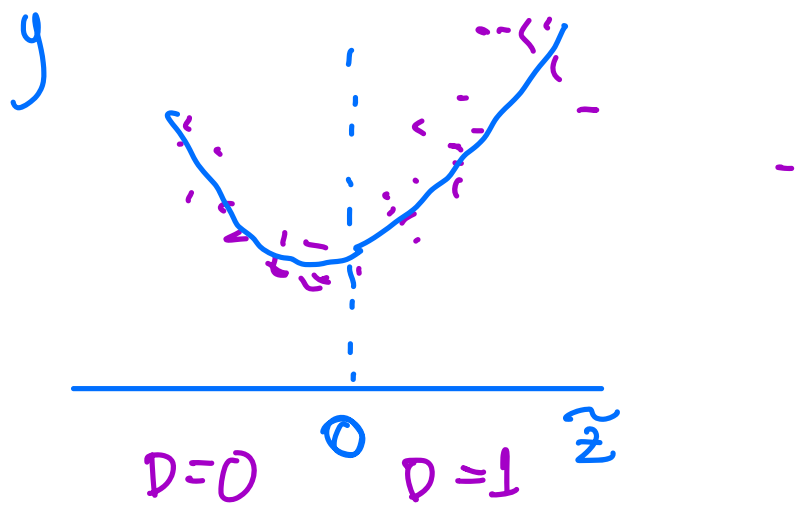
$$\tau(z) = E[y(D=1) - y(D=0) | z] = (\delta_1 - \delta_0) + (\beta_1 - \beta_0)z$$

$$E[y | z, D] = D \cdot E[y(D=1) | z] + (1-D) E[y(D=0) | z]$$

$$= (\delta_1 - \delta_0) + \tau D + \beta_0 \tilde{z} + \beta_1 D \times \tilde{z}$$



$$E[y | z, D] = \delta + \tau D + \beta_0 \tilde{z} + \beta_1 D \cdot \tilde{z} + \beta_2 \tilde{z}^2 + \beta_3 \tilde{z}^2 \cdot D$$



- Pruebas de Robustez

1 Sensibilidad: ¿Qué tan sensibles son sus resultados a cambios en la especificación?

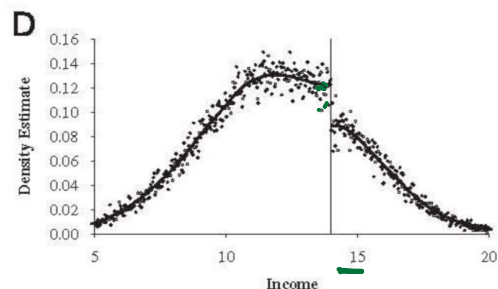
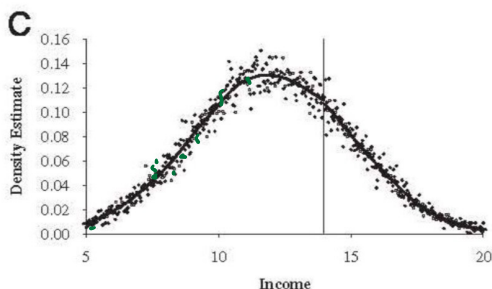
2 Balance: ¿Alguna de las variables de control salta en el umbral?

- Haga pruebas gráficas

$$E[X | Z, D] = \beta_0 + \beta_1 \tilde{Z} + \tau D + \tilde{\beta}_3 \tilde{Z} \cdot D$$

3 Placebo: ¿se umbrales placebo $\tilde{Z}$

4 Sorting: ¿Los participantes cambian su comportamiento en el umbral?



Test McCrary 2008.

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Regresión Discontinua Borrosa

z_i es la variable de asignación

c es el umbral.

$$w_i = 1[z_i > c] \quad (1[z_i < c])$$

Supuestos de Identificación

- 1 Continuidad Local $D(w=1), D(w=0), y(w=1), y(w=0)$ son continuos alrededor del umbral.
- 2 Supuestos de Variables instrumentales: monotonización, restricción de exclusión y relevancia de z .

RDB $\Rightarrow$ LOCAL² ATE (cooperadores en el umbral)

$$\tau_{RDB} = E[y(w=1) - y(w=0) \mid \text{Unidad } i \text{ es un cooperador \& } z_i = \bar{z}]$$

$$\hat{\tau}_{RDB} = \frac{\lim_{z \downarrow \bar{z}} E[y | z = \bar{z}] - \lim_{z \uparrow \bar{z}} E[y | z = \bar{z}]}{\lim_{z \downarrow \bar{z}} E[D | z = \bar{z}] - \lim_{z \uparrow \bar{z}} E[D | z = \bar{z}]}$$

$$= \frac{\text{Discontinuidad en } y}{\text{Discontinuidad en } D}$$

$$\hat{\tau}_{\text{LSD}} = \frac{E[Y|W=1] - E[Y|W=0]}{E[D|W=1] - E[D|W=0]} \quad \checkmark$$

Estimación



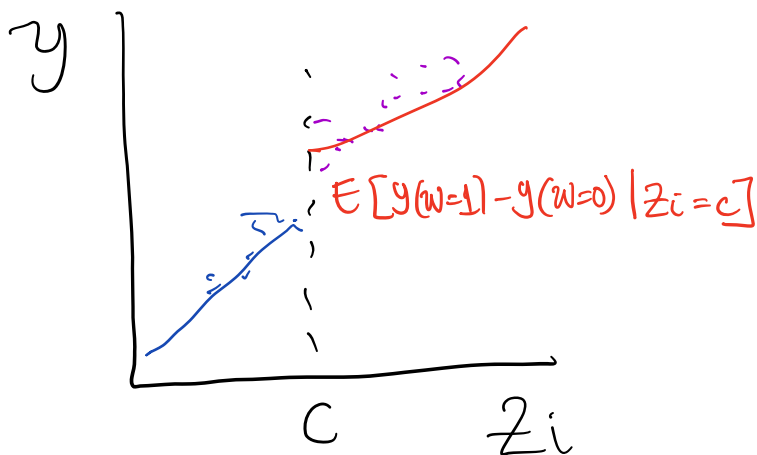
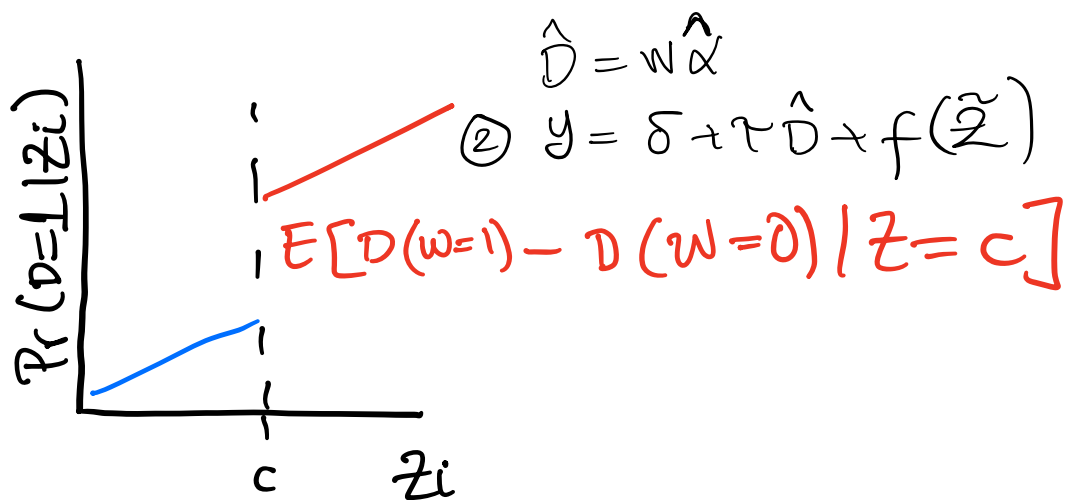
0. Centrar la variable $\tilde{Z} = Z - \bar{Z}$

1. Seleccione la muestra (ancho de Banda)

2. Cree el instrumento $W = 1[Z > \bar{Z}]$

3. Estime los MCOZE. $Y = \delta + \tau D + f(\tilde{Z})$

① $D = W\alpha + u \rightarrow W$ es el instrumento.



How to choose this bandwidth:

Using a larger bandwidth yields more precise estimates as more observations are available to estimate the regression. But it can bias the estimate since you are comparing ind that are very different

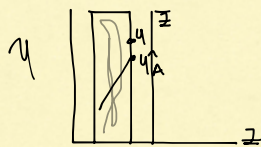
In econometrics (& statistics) two procedures are generally considered for choosing bandwidths.

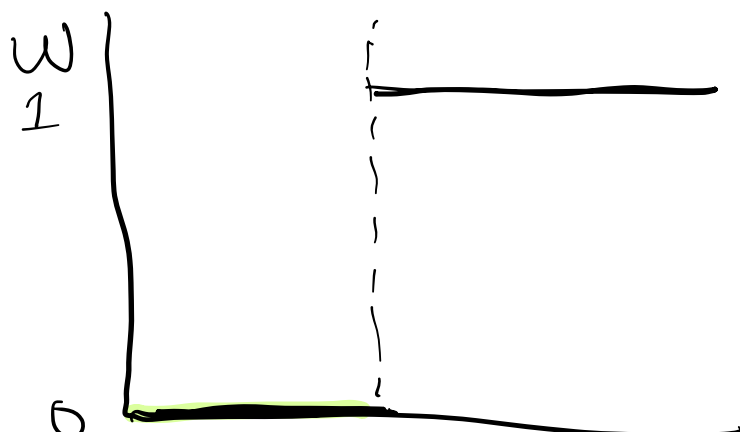
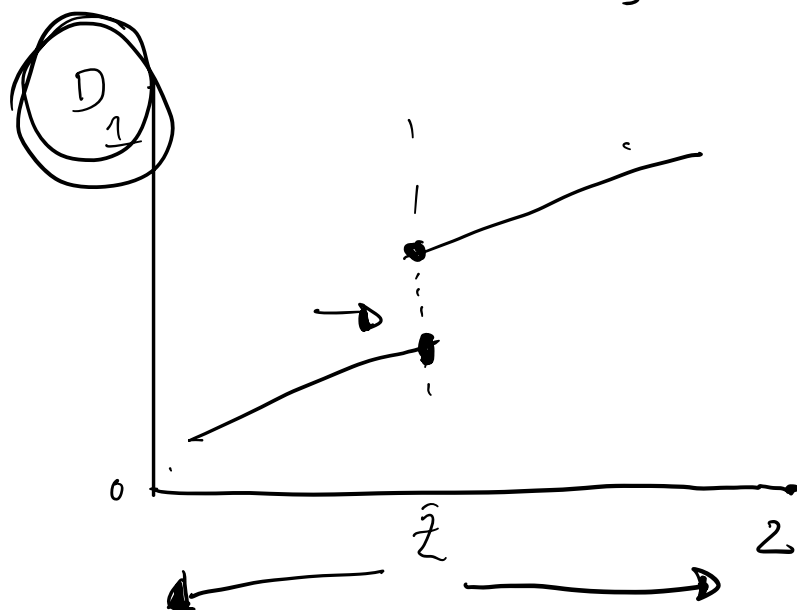
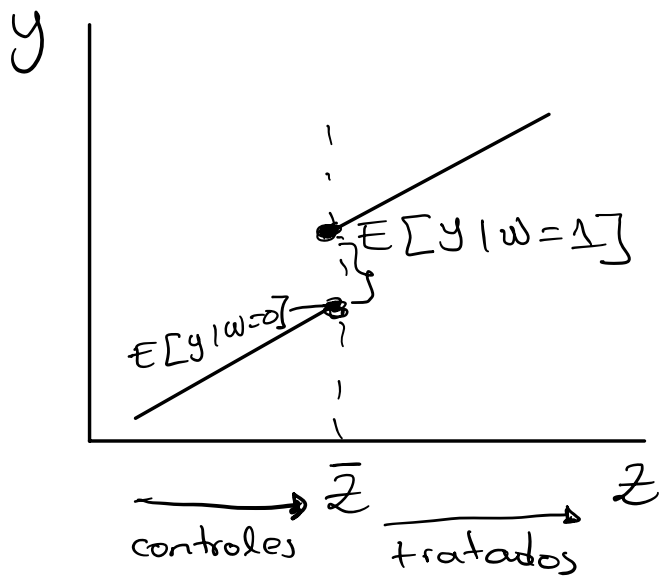
1. Characterizing the optimal bandwidth in terms of the unknown joint distribution of all variables
2. Cross validation procedure (estimating the regression function at the boundary)

Consider observation i

- To see how well a linear regression with a bandwidth h fits the data, we run a reg with obs i left out and use the estimates to predict the value of y at $x = x_i$
- RD: The reg is estimated using only obs with values of Z on the left of Z_i

1. Select a bandwidth h_1
2. Start with an observation A to the left of the wt-point, with rating z_A & y_A
3. To see how well the parametric assumption fits the data within the bandwidth h_1 , run a regression of the outcome on the rating using all of the observations that are located to the left of observation A and have a rating that ranges from $z_A - h_1$ to z_A (Not including z)
4. Get the predicted value of the outcome Variable observation A based on this regression and call it $\hat{y}_A$
5. Shift the "band" slightly over to the left and repeat this process to obtain predicted values for B . Repeat this process to obtain predicted values for all observations to the left of the wt-point
6. Repeat to the right
7. Calculate
$$CV(h_1) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$
8. Repeat for other bandwidth choices
9. Pick the bandwidth that minimizes the cross-validation criterion





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$\bar{z}$

$\overline{z}$